

Qiuyan Dong

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EDUCATION

University of California, Santa Barbara

Bachelor of Science in Statistics and Data Science

Santa Barbara, CA

09/2024 - 06/2027

- Cumulative GPA: 3.86/4.00
- Relevant Coursework: Linear Algebra, Vector Calculus, Differential Equations, Probability and Statistics, Stochastic Process, Computer Science, Intermediate Python, Data Science Principles
- Honors and Awards: UCSB Dean's Honors (Fall 2024, Spring 2025)

RESEARCH/PROJECTS EXPERIENCE

Land Use Optimization Modeling for Sustainable Development

07/2025-08/2025

- Developed and integrated a three-stage analytical pipeline for optimal land-use planning, combining a computer vision-based clustering model (OpenCV) for spatial segmentation, a multi-criteria evaluation framework, and a genetic algorithm for optimization.
- Designed a multi-factor evaluation model that quantified economic, environmental, and community impacts; applied Analytic Hierarchy Process (AHP) to determine weights and transform subjective stakeholder priorities into objective, quantifiable metrics.
- Implemented a genetic algorithm in Python to solve the resulting optimization problem, searching a large solution space to identify building configurations that maximized composite benefits, and validated outcomes through sensitivity and generalizability analysis.

Species Classification for Lizard Biodiversity Analysis

04/2025-06/2025

- Completed a full-cycle data science project utilizing Python and scikit-learn to implement and compare three classification models (SVM, AdaBoost, Random Forest) for distinguishing 8 lizard species and sex based on morphometric data.
- Engineered SVM models with tailored kernel functions and ensemble techniques, achieving up to 99.82% accuracy in species isolation and improving integrated model performance through feature optimization and noise elimination.
- Improved model robustness by conducting sensitivity analysis and 8-fold cross-validation, and built a preprocessing pipeline with StandardScaler and PCA for dimensionality reduction and improved generalizability.

Climate Change Modeling: CO₂ Concentration & Global Temperature Correlation

01/2024 – 03/2025

- Developed and validated multiple predictive models, including regression, ARIMA, and BP neural network, to analyze and forecast atmospheric CO₂ concentration levels and global mean surface-air temperature trends.
- Built and fine-tuned ARIMA time-series models for CO₂ and temperature, achieving a 0.999 R-square fit; enhanced the regression model's reliability by integrating ARIMA to fit residuals and validating model superiority through systematic comparison with model baselines.
- Conducted a comprehensive sensitivity analysis on key parameters to ensure robustness of ARIMA and BP neural network models.
- Established a strong linear correlation between CO₂ and temperature with Pearson and Spearman coefficients of 0.961 and 0.949, and translated model findings into data-driven policy recommendations for emission reduction.

Intelligent Course Scheduling System Design Using Genetic Algorithm

10/2022 – 03/2023

- Led a full-cycle data science project from initial user interviews with 4+ school administrators to defining problem scopes, resulting in a comprehensive requirement framework with 4 hard and 8 soft scheduling constraints for modeling.
- Formulated a multi-objective optimization model with a customized fitness function that integrated key metrics, including classroom utilization, individual walking distance, and course timing preferences, to quantitatively evaluate schedule quality.
- Engineered a genetic algorithm from scratch in Python, implementing adaptive crossover/mutation probabilities and conflict-resolution logic; validated its efficacy through a pilot test (4 classes, 5 daily sessions), significantly optimizing scheduling efficiency and fully resolving movement conflicts.
- Developed a prototype mini-program interface to visualize optimized schedules and support scenario testing, and drafted an implementation roadmap for future school-wide deployment and A/B testing.

PROFESSIONAL EXPERIENCE

Marine Engineering Environmental Research Center, Ministry of Natural Resources

Qingdao, China

Digitalization and Data Analyst Intern

07/2025 - 08/2025

- Participated in consolidating disparate research, experimental, and marine monitoring systems into a unified data lakehouse; assisted in standardizing metadata documentation and establishing SQL-accessible retrieval paths for streamlined data access.
- Developed dynamic data visualization dashboards using Python and JavaScript modules by building ETL pipelines that integrated real-time satellite, sensor, and survey data, enabling continuous monitoring of hydrological and ecological conditions.
- Applied machine learning models, including XGBoost, LSTM, and CNN, for marine climate simulation and extreme event prediction; optimized model performance through cross-validation and regularization, and validated robustness via backtesting, achieving improved AUC and prediction accuracy.
- Collaborated on developing optimization models leveraging linear programming to support decision-making in coastal island resource planning, balancing development objectives with environmental constraints to generate actionable recommendations.

RuiXue Education

Qingdao, China

Teaching Assistant and Science Tutor

06/2024 - 09/2024

- Tutored 50+ K–12 students while building a data analytics pipeline in Python to track performance, identify learning gaps, and personalize instruction, contributing to a 15% average score improvement.
- Assisted in developing a recommendation engine integrating statistical models and a large-language-model interface to generate adaptive problem sets and instant feedback, enhancing individualized learning experiences.

OTHER INFORMATION

Programming and Computer Science Skills: Python (Pandas, Scikit-learn, PyTorch, Plotly), Matlab, Java, C++

Languages: Chinese (Native), English (Proficient)

Interests and Extracurricular Activities: Pipa (Chinese lute), Traditional Chinese Painting